

Product Requirements Document (PRD)

PixelVault

Online Gaming Website

1. Product Overview

PixelVault is a browser-based indie game discovery and publishing platform designed to connect players with independent game creators. The platform enables users to discover games through curated content, explore detailed game information, save games to collections, submit ratings and reviews, and manage activity through personal profiles.

The platform also provides a dedicated developer area where creators can publish games, manage listings, and monitor engagement metrics. PixelVault focuses on game discovery, community participation, and creator visibility through structured content organization and featured promotion.

2. Problem Statement

Players interested in browser-based indie games often rely on multiple websites and communities to discover new titles. This fragmented experience makes game discovery inconsistent and reduces visibility for smaller developers.

Independent developers face challenges in promoting browser-based games and reaching players interested in discovering new content. Existing gaming platforms frequently prioritize larger publishers and commercial releases.

PixelVault addresses these challenges by providing a centralized environment for discovering, publishing, reviewing, and organizing browser-based indie games.

3. Product Goals

Discovery Goals

- Enable users to discover browser-based indie games through curated content.
- Promote featured and trending games.
- Support game exploration through categories, search, and filtering.

Engagement Goals

- Encourage users to create and manage game collections.
- Increase participation through ratings and reviews.
- Support ongoing engagement through profile activity and saved content.

Creator Goals

- Provide developers with a structured publishing workflow.
- Support management of game listings.
- Provide visibility into game engagement metrics.

Platform Goals

- Maintain an organized game catalog.
 - Highlight selected games through featured content areas.
 - Support moderation of platform content.
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4. Target Audience

Casual Players

Users who want to discover and play browser-based games without downloading software.

Indie Game Enthusiasts

Users interested in exploring independent game releases and discovering new creators.

Independent Developers

Creators publishing browser-based games who require a dedicated platform for visibility and engagement.

Community Contributors

Users who participate through collections, ratings, reviews, and recommendations.

5. User Personas

Casual Explorer

- Browses featured content.
- Discovers games through categories.
- Saves games for future sessions.
- Submits ratings after gameplay.

Dedicated Collector

- Creates game collections.
- Tracks recently played games.
- Shares reviews and recommendations.
- Uses profile activity features.

Indie Developer

- Publishes browser-based games.
 - Maintains game information.
 - Reviews engagement data.
 - Updates listings over time.
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6. User Journey

Player Journey

1. Create an account.
2. Browse homepage content.
3. Explore games through search and categories.
4. Open a game detail page.
5. Launch a game.
6. Save the game to a collection.
7. Submit a rating and review.
8. Access profile activity and collections.

Developer Journey

1. Create a developer account.
 2. Access the developer dashboard.
 3. Submit a game listing.
 4. Add game information and media assets.
 5. Publish the game.
 6. Review engagement metrics.
 7. Manage existing listings.
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7. Technology Requirements

The platform must be developed using:

- React
- TypeScript

- Tailwind CSS
- React Router
- JSON-based local data storage
- Local Storage for session persistence

The application will function as a frontend-focused web application and will not require a backend service.

The platform must support desktop and mobile browsers.

8. Reference Assets

The project includes reference assets that define the intended direction for:

- Homepage layout
- Game library layout
- Game detail page layout
- User profile layout
- Developer dashboard layout
- Navigation structure
- Branding direction
- Design system

The reference assets are intended to guide structure, content hierarchy, and visual direction. The assets should not be reproduced exactly.

9. Core Product Modules

Homepage

The homepage serves as the primary discovery hub and includes:

- Hero banner
 - Featured Games section
 - Trending Games section
 - Game of the Week section
 - Genre browsing section
 - Navigation menu
 - Footer section
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Game Library

The game library provides:

- Search functionality
 - Genre filtering
 - Sorting controls
 - Game card grid layout
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Game Detail Experience

Each game detail page provides:

- Game banner
 - Game information
 - Genre information
 - Screenshot gallery
 - Ratings summary
 - Reviews section
 - Play Game action
 - Developer information
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User Profiles

User profiles provide:

- Account information
 - Recently played games
 - Saved collections
 - Review history
 - Activity overview
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Collections

Users can:

- Create collections
 - Save games
 - Remove games from collections
 - View collection contents
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Ratings and Reviews

Users can:

- Submit ratings
 - Submit reviews
 - View community feedback
 - Browse review history
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Developer Portal

Developers can:

- Create game listings
 - Edit game listings
 - Publish game listings
 - Manage existing games
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Developer Analytics

Developers can view:

- Total Games
 - Total Downloads
 - Total Revenue
 - Followers
 - Analytics Overview Graph
 - Recent Activity
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Featured Content Management

The platform supports:

- Featured Games
 - Trending Games
 - Game of the Week
 - Curated content sections
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Administration

Administrative functions include:

- Game moderation
 - Review moderation
 - Featured content management
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10. Functional Requirements

Discovery

- Users can browse featured games.
- Users can browse trending games.
- Users can browse games by genre.
- Users can search for games.
- Users can filter available games.
- Users can sort available games.

Game Experience

- Users can access game detail pages.
- Users can view screenshots.
- Users can view ratings.
- Users can view reviews.
- Users can launch browser-based games.

User Accounts

- Users can create accounts.
- Users can log in.
- Users can access profiles.
- Users can manage collections.
- Users can view activity history.

Community

- Users can submit ratings.
- Users can submit reviews.
- Users can view community feedback.

Developer Features

- Developers can create listings.
- Developers can edit listings.
- Developers can publish listings.

- Developers can access analytics information.

Administration

- Administrators can moderate games.
 - Administrators can moderate reviews.
 - Administrators can manage featured content.
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11. Non-Functional Requirements

- The platform supports desktop and mobile browsers.
 - Navigation remains consistent across pages.
 - User activity remains associated with the correct account.
 - Collections remain associated with the correct user.
 - Ratings remain associated with the correct game.
 - Reviews remain associated with the correct game.
 - Published games remain accessible through the game library.
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12. Success Metrics

The platform may be evaluated using:

- Number of registered users.
 - Number of published games.
 - Number of saved collections.
 - Number of ratings submitted.
 - Number of reviews submitted.
 - Number of game detail page visits.
 - Number of developer accounts.
 - Number of active game listings.
 - Number of featured content interactions.
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13. In Scope

- Homepage
- Game Library
- Search and filtering
- Game Detail Pages
- User Authentication
- User Profiles
- Collections

- Ratings and Reviews
 - Developer Dashboard
 - Game Submission Workflow
 - Developer Analytics
 - Featured Content Management
 - Administrative Moderation
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14. Out of Scope

- Payment processing
 - Subscription plans
 - Digital purchases
 - Multiplayer functionality
 - Live chat
 - Voice communication
 - Live streaming
 - Tournament systems
 - Third-party advertising
 - Native desktop applications
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15. Submission Requirements

The final submission must include:

- Complete source code
- README documentation
- Dockerfile
- Package configuration files
- Static data files
- Project assets
- Environment example file
- Instructions for local setup and execution